

The Variational Quantum Eigensolver

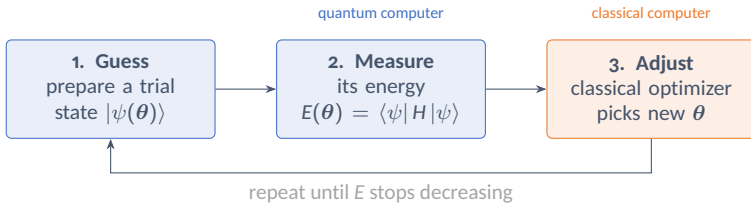
Ground states by optimization — worked end to end

Lecture 6

September 29, 2026

The variational idea

- **Goal:** the smallest eigenvalue E_0 of a Hamiltonian H on n qubits
- **Direct route:** diagonalize a $2^n \times 2^n$ matrix — hopeless beyond $n \approx 20$
- **Variational route:** don't solve for the answer — *search* for it



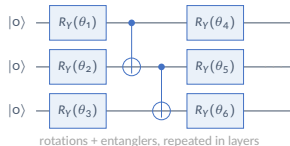
Takeaway: Like tuning the knobs of a radio: you never compute the right setting, you turn the knobs and keep what sounds better. Here “sounds better” means *lower energy*.

The ansatz — a circuit with knobs

- Ansatz is a **parameterized circuit**

$$|\psi(\theta)\rangle = U(\theta) |0\rangle^{\otimes n}$$

- The angles θ are the knobs
- Try <https://pqc-wqd.streamlit.app/>
- It fixes **which states the search can reach**
— VQE finds the best state *inside* this family, nothing outside it



Two design styles

- **Hardware-efficient** (above): native gates, shallow — but too expressive $\Rightarrow$ barren plateaus (L5)
- **Problem-inspired**: built from the structure of H — few parameters, right subspace (our H_2 example)

Measurement — turning a state into a number

Hardware cannot measure “energy”. So write H as a sum of **Pauli strings**:

$$H = \sum_P c_P P \quad \implies \quad E(\theta) = \sum_P c_P \langle P \rangle$$

- Each shot gives ± 1 ; $\langle P \rangle$ is the average
- Hardware reads only Z — rotate first:

To measure	apply first
Z	nothing
X	H
Y	$S^\dagger$, then H

- $Z_0 Z_1$: measure both, multiply the two ± 1 outcomes
- Terms needing the same rotations **share shots**
- **Cost:** error $\sim 1/\sqrt{N} - 10^{-3}$ precision needs $\sim 10^6$ shots

Takeaway: The quantum computer outputs ± 1 samples; the classical side combines them into $E(\theta)$.

Background — eigenvalues of a Hermitian matrix

- An eigenvector $|\psi_i\rangle$ of H satisfies

$$H |\psi_i\rangle = \lambda_i |\psi_i\rangle$$

and H has $N = 2^n$ eigenvalues $\lambda_1, \dots, \lambda_N$

- We want the smallest one: $\lambda_{\min} = \min(\lambda_1, \dots, \lambda_N)$
- Every Hamiltonian is **Hermitian**, $H = H^\dagger$, which buys two things:
 - all λ_i are *real* — they are measurable energies
 - the $|\psi_i\rangle$ form an orthonormal basis
- **Spectral theorem:**

$$H = \sum_{i=1}^N \lambda_i |\psi_i\rangle \langle \psi_i|$$

- For a Hamiltonian we also write $\lambda_{\min} = E_0$, the ground-state energy

Takeaway: Diagonalizing H answers the question exactly — but H is a $2^n \times 2^n$ matrix, which is why we look for $\lambda_{\min}$ instead of computing it.

Example

$$H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = X$$

$$\lambda_+ = +1 : |+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$\lambda_- = -1 : |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

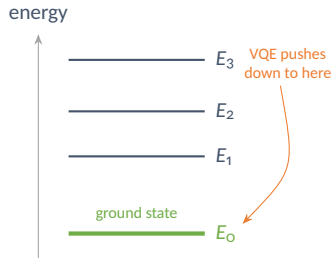
So $\lambda_{\min} = -1$, with ground state $|-\rangle$, and $H = |+\rangle \langle +| - |-\rangle \langle -|$.

The lowest eigenvalue is the ground state

- H is the **energy operator**; its eigenvalues are the only energies the system can have

$$H |E_k\rangle = E_k |E_k\rangle, \quad E_0 \leq E_1 \leq E_2 \leq \dots$$

- **Ground state** $|E_0\rangle$: the eigenvector with the lowest eigenvalue
- Why we want it:
 - **chemistry**: E_0 is the molecule's energy
 - **optimization**: we *build* H so that $|E_0\rangle$ is the best solution (L4)



The variational principle — why minimizing works

- Write *any* trial state in the eigenbasis: $|\psi\rangle = \sum_k c_k |E_k\rangle$, $\sum_k |c_k|^2 = 1$

$$\langle\psi|H|\psi\rangle = \sum_k |c_k|^2 E_k \geq \sum_k |c_k|^2 E_0 = E_0$$

- The measured energy is a **weighted average of eigenvalues** — it can never go below the smallest one
- Equality only when all the weight sits on $|E_0\rangle$, i.e. $|\psi\rangle = |E_0\rangle$

Consequence

$\min_{\theta} E(\theta) \geq E_0$ for *any* ansatz. Every VQE result is an **upper bound**: lower is always better, and a poor ansatz or noise can only push the number *up*.

The objective — minimize an expectation value

Every trial state obeys $\lambda_{\theta} \equiv \langle \psi(\theta) | H | \psi(\theta) \rangle \geq \lambda_{\min}$, so the best we can do is tighten that bound.

The VQE problem

$$\min_{\theta} \langle \psi(\theta) | H | \psi(\theta) \rangle = \lambda_{\theta^*}$$

$$\lambda_{\min} \leq \lambda_{\theta^*} \approx E_{\text{gs}}$$

- Minimizing an *expectation value* replaces diagonalizing H
- Equality needs $|\psi(\theta)\rangle = |\psi_{\min}\rangle$: the ansatz decides how tight the bound can get

Toy check: $H = X$

Ansatz $|\psi(\theta)\rangle = R_y(\theta) |0\rangle$:

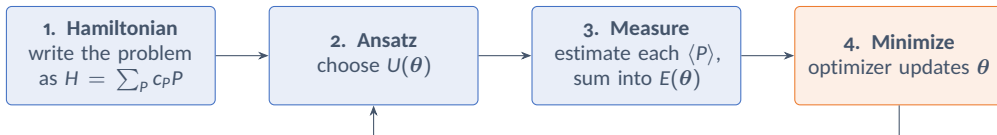
$$E(\theta) = \langle \psi | X | \psi \rangle = \sin \theta$$

$$\min_{\theta} E(\theta) = -1 \text{ at } \theta^* = -\frac{\pi}{2}$$

and $|\psi(\theta^*)\rangle = |-\rangle$ — both $\lambda_{\min}$ and its eigenvector.

Why not phase estimation? Exact, but it needs controlled powers of e^{-iHt} — a long coherent circuit, so fault-tolerant hardware.

The VQE recipe — used for both examples



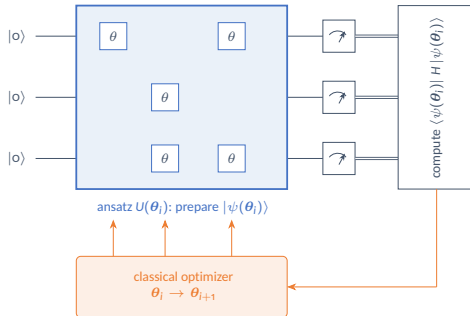
	Example 1: hydrogen molecule	Example 2: Max-Cut
H comes from	physics (electrons and nuclei)	the problem (edges of a graph)
Answer we want	the lowest <i>energy</i>	the lowest-energy <i>bit string</i>

Takeaway: Step 1 is where the problem enters. Everything after it is the same machine.

The VQE loop, at circuit level

One iteration

- 1 Prepare $|\psi(\theta_i)\rangle$
- 2 Measure, one Pauli setting at a time
- 3 Combine counts into $\langle \psi(\theta_i) | H | \psi(\theta_i) \rangle$
- 4 Optimizer proposes θ_{i+1}



The circuit is re-run *from scratch* every iteration — nothing stays coherent between steps.

Takeaway: The quantum side answers one question per iteration — “what is the energy of *this* state?” The classical side decides which state to ask about next.

Example 1 — The hydrogen molecule (H_2)

From a molecule to a 2-qubit Hamiltonian

H₂: how the Hamiltonian is formed



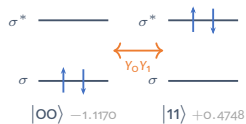
- 1 **Physics:** kinetic energy + electron-nucleus attraction + electron-electron and nucleus-nucleus repulsion
- 2 **Orbitals:** $\sigma = 1s_A + 1s_B$, $\sigma^* = 1s_A - 1s_B$. A *classical* code (PySCF) computes the energy integrals between them
- 3 **To qubits:** $|1\rangle$ = this spin-orbital holds an electron $\Rightarrow$ 4 qubits
- 4 **Shrink:** there is always one $\uparrow$ and one $\downarrow$ electron, so qubit 0 = orbital of the $\uparrow$ electron ($|0\rangle = \sigma$, $|1\rangle = \sigma^*$), qubit 1 = same for $\downarrow$

H₂: the Hamiltonian, term by term

$$H = -0.3324 I - 0.3979 Z_0 - 0.3979 Z_1 + 0.0113 Z_0 Z_1 - 0.1809 Y_0 Y_1 \quad (\text{Ha})$$

Term	Meaning
I	constant (includes nuclear repulsion)
Z_0, Z_1	cost of lifting the $\uparrow / \downarrow$ electron to σ^*
$Z_0 Z_1$	small repulsion correction
$Y_0 Y_1$	moves both electrons $\sigma \leftrightarrow \sigma^*$ — the only <i>off-diagonal</i> term

- Z terms give each configuration's energy; $Y_0 Y_1$ **mixes** them
- Without $Y_0 Y_1$ the answer is just $|00\rangle$ (Hartree-Fock)



In the $\{|00\rangle, |11\rangle\}$ block:

$$\begin{pmatrix} -1.1170 & 0.1809 \\ 0.1809 & 0.4748 \end{pmatrix}$$

H₂: ansatz, measurement, result

Ansatz — one knob:



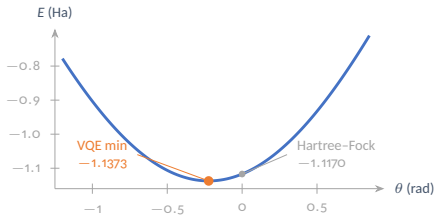
Measure — two settings:

- Z basis: $\langle Z_0 \rangle = \langle Z_1 \rangle = \cos \theta$, $\langle Z_0 Z_1 \rangle = 1$
- Y basis: $\langle Y_0 Y_1 \rangle = -\sin \theta$

Combine:

$$E(\theta) = -0.3211 - 0.7959 \cos \theta + 0.1809 \sin \theta$$

Takeaway: $\theta^* = -12.8^\circ$ gives $E = -1.1373$ Ha, the exact answer in this basis. The 20.3 mHa below Hartree-Fock is the **correlation energy** ($\gg 1.6$ mHa chemical accuracy).



Example 2 — Max-Cut

From a graph to an Ising Hamiltonian

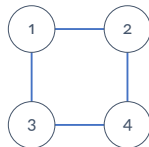
Max-Cut: how the Hamiltonian is formed

- 1 **Encode:** one qubit per vertex, $|0\rangle / |1\rangle =$ side of the cut.
 Z_i reads it as $z_i = \pm 1$
- 2 **Score an edge:** cut $\iff z_i \neq z_j \iff z_i z_j = -1$

$$\text{cut}_{ij} = \frac{1}{2}(1 - z_i z_j) \in \{0, 1\}$$

- 3 **Sum over edges** and flip sign to minimize; constants drop out:

$$H_C = \sum_{(i,j) \in E} w_{ij} Z_i Z_j, \quad \text{cut} = \frac{1}{2} \left(\sum w_{ij} - E \right)$$



The L3 square, all $w_{ij} = 1$:

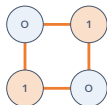
$$H_C = Z_1 Z_2 + Z_1 Z_3 + Z_2 Z_4 + Z_3 Z_4$$

each edge: +1 if uncut, -1 if cut

A. Lucas (2014). "Ising formulations of many NP problems." *Frontiers in Physics* 2, 5. arXiv:1302.5843

Max-Cut: the eigenvalues are the cut values

- Only Z s $\Rightarrow H_C$ is **diagonal**
- Every bit string is an eigenstate; its eigenvalue scores that cut
- **Ground state = maximum cut**
- $|0110\rangle$ and $|1001\rangle$: the same cut, seen from either side



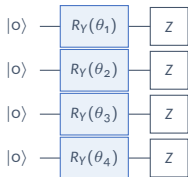
$|0110\rangle$: all 4 cut

Bit string	cut	E	#
0000, 1111	0	+4	2
e.g. 1000, 1100	2	0	12
0110, 1001	4	-4	2

Contrast with H_2

H_2 's ground state is a *superposition* (because of $Y_0 Y_1$). Here it is a *single bit string*.

Max-Cut: ansatz, measurement, result



one rotation per vertex; no
entanglement needed

- **Measure:** one setting, all qubits in Z
- $\langle Z_i Z_j \rangle = \cos \theta_i \cos \theta_j$ for this product state, so

$$E(\theta) = c_1 c_2 + c_1 c_3 + c_2 c_4 + c_3 c_4, \quad c_i = \cos \theta_i$$

- **Minimize:** $\theta^* = (0, \pi, \pi, 0) \Rightarrow |0110\rangle$, $E = -4$, cut = 4
- **Keep the best sample, not the average:** every shot is a cut you can check classically
- On harder graphs this simple ansatz gets stuck $\Rightarrow$ L7 derives a better one from H_C (QAOA)

The two examples side by side

	H ₂ molecule	Max-Cut (square)
<i>H</i> built from	physics, via orbital integrals	the edges, $\frac{1}{2}(1 - z_i z_j)$ each
A qubit means	which orbital an electron occupies	which side a vertex is on
Pauli terms	$I, Z_0, Z_1, Z_0 Z_1, Y_0 Y_1$	$Z_i Z_j$ only
<i>H</i> diagonal?	no — $Y_0 Y_1$ mixes states	yes
Ground state	a superposition (entangled)	a single bit string
Ansatz	$R_Y(\theta) + \text{CNOT}$, 1 parameter	$R_Y(\theta_i)$ per qubit, 4 params
Measurement settings	2 (Z basis, Y basis)	1 (Z basis)
What we keep	the energy $E(\theta^*)$	the best sampled bit string

Takeaway: Same algorithm, different Step 1. The Hamiltonian decides how hard measurement is and what kind of answer you read out.

Where VQE struggles

- **Measurement cost** — $\sim L/\epsilon^2$ shots per iteration; molecules have $L \sim n^4$ Pauli terms
- **Barren plateaus** (L5) — expressive ansätze have vanishing gradients as n grows
- **Local minima** — the landscape is non-convex, as the Max-Cut caveat showed
- **Noise** — a noisy device optimizes a *different* cost; its minimum need not be near E_0

■ *Our two examples were solvable by hand on purpose. For H_2 , the basis-set error (true energy ≈ -1.174 Ha) is larger than anything VQE fixes. Keep that in view when a paper reports a small-molecule success.*

J. Tilly et al. (2022). “The Variational Quantum Eigensolver: a review of methods and best practices.” *Physics Reports* 986, 1–128. arXiv:2111.05176